

Centre of Mass, Linear Momentum and Collisions

Systems of particles: the point that moves as if all the mass were there, and the quantity that no internal force can change.

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So far we have treated every body as a single particle. Real systems contain many interacting parts: a spinning wrench, an exploding firework, two colliding cars, a rocket throwing out exhaust. Two ideas make such systems manageable. The **centre of mass** moves as if all the mass were concentrated there and all the external forces acted on it. **Linear momentum** is conserved whenever the external forces add to zero, however violent the internal forces. Together they let us analyse collisions without knowing anything about the messy forces during impact. That is exactly the situation in crash testing, sports science, particle physics and rocketry.

YOU WILL BE ABLE TO

- Locate the centre of mass of discrete and continuous systems, including bodies with holes.
- Show that the centre of mass moves as a particle acted on by the net external force.
- Use impulse and the impulse–momentum theorem to relate forces, contact times and changes of velocity.
- Apply conservation of momentum to recoil, explosions and collisions in one and two dimensions.
- Classify collisions as elastic, inelastic or perfectly inelastic, and use the coefficient of restitution.
- Derive and apply the rocket equation for variable-mass systems.

THINK FIRST

You are floating at rest in the middle of a frictionless ice rink, holding a heavy textbook. How can you get to the edge?

Throw the book. You and the book start with zero total momentum, and the ice exerts no horizontal force, so the total momentum must stay zero. If you throw the book one way, you must move the other way, with $m_{\text{you}}v_{\text{you}} = m_{\text{book}}v_{\text{book}}$. Waving your arms or wriggling does not help: internal forces cannot move the centre of mass. This is exactly how rockets work. They throw mass (exhaust) backwards in order to move forwards, with no need for anything to “push against”.

§6.1

The centre of mass

For a system of particles with masses m_i at positions $\vec{r}_i$, the **centre of mass** (CM) is the mass-weighted average position:

$$\vec{r}_{\text{cm}} = \frac{\sum m_i \vec{r}_i}{M}, \quad M = \sum m_i.$$

In components, $x_{\text{cm}} = \sum m_i x_i / M$, and similarly for y and z .

For a continuous body, the sum becomes an integral over mass elements dm :

$$\vec{r}_{\text{cm}} = \frac{1}{M} \int \vec{r} dm.$$

For a rod, $dm = \lambda dx$, where λ is the mass per unit length. For a plate, $dm = \sigma dA$. For a solid, $dm = \rho dV$.

Useful facts:

- If a body has a line or plane of symmetry and uniform density, the CM lies on it.
- The CM need not lie inside the material: think of a ring, a horseshoe or a boomerang.
- To find the CM of a composite body, treat each part as a particle at its own CM. A **hole** can be treated as a part with *negative* mass.
- In a uniform gravitational field, the CM coincides with the **centre of gravity**, the point where the total weight effectively acts.

WORKED EXAMPLE 6.1

Three particles

Masses of 1.0 kg, 2.0 kg and 3.0 kg sit at (0, 0), (2.0, 0) and (0, 4.0) m. Find the centre of mass.

Solution

$$x_{\text{cm}} = \frac{1(0) + 2(2.0) + 3(0)}{6.0} = \frac{4.0}{6.0} = 0.67 \text{ m}, \quad y_{\text{cm}} = \frac{1(0) + 2(0) + 3(4.0)}{6.0} = 2.0 \text{ m}.$$

The CM is at (0.67, 2.0) m, pulled towards the heaviest mass.

WORKED EXAMPLE 6.2

Non-uniform rod and semicircular wire

(a) A rod of length L lies along the x -axis from 0 to L . Its density increases linearly from one end to the other: $\lambda(x) = \lambda_0(1 + x/L)$. Find its centre of mass. (b) Find the CM of a thin uniform wire bent into a semicircle of radius R .

Solution

(a) Find the total mass and the first moment:

$$M = \int_0^L \lambda_0 \left(1 + \frac{x}{L}\right) dx = \lambda_0 \left(L + \frac{L}{2}\right) = \frac{3}{2} \lambda_0 L.$$

$$\int_0^L x \lambda dx = \lambda_0 \int_0^L \left(x + \frac{x^2}{L}\right) dx = \lambda_0 \left(\frac{L^2}{2} + \frac{L^2}{3}\right) = \frac{5}{6} \lambda_0 L^2.$$

$$x_{\text{cm}} = \frac{\frac{5}{6} \lambda_0 L^2}{\frac{3}{2} \lambda_0 L} = \frac{5}{9} L \approx 0.556 L.$$

This is beyond the midpoint, towards the denser end, as expected.

(b) Put the centre of the semicircle at the origin, with the wire in the upper half-plane. By symmetry $x_{\text{cm}} = 0$. Parametrise by the angle θ from 0 to π : a short piece of arc has length $R d\theta$ and mass $dm = \lambda R d\theta$, at height $y = R \sin \theta$.

$$y_{\text{cm}} = \frac{1}{\lambda \pi R} \int_0^\pi (R \sin \theta) \lambda R d\theta = \frac{R}{\pi} [-\cos \theta]_0^\pi = \frac{2R}{\pi} \approx 0.64 R.$$

The CM lies in empty space, inside the arc but not on the wire. By a similar calculation, a uniform semicircular plate (half-disc) has $y_{\text{cm}} = 4R/3\pi \approx 0.42 R$. It is lower, because more of the plate's mass lies near the flat edge.

WORKED EXAMPLE 6.3

A disc with a hole

A circular hole of radius $R/2$ is cut from a uniform disc of radius R . The hole touches the edge of the disc, so its centre is at distance $R/2$ from the disc's centre. Where is the CM of the remaining plate?

Solution

Put the origin at the disc's centre, with the hole centred at $x = +R/2$. Mass is proportional to area, so if the full disc has mass M , the removed piece has mass $M/4$ (a quarter of the area).

Treat the plate as “full disc” plus “hole of negative mass”:

$$x_{\text{cm}} = \frac{M(0) + \left(-\frac{M}{4}\right)\left(\frac{R}{2}\right)}{M - \frac{M}{4}} = \frac{-MR/8}{3M/4} = -\frac{R}{6}.$$

The CM shifts by $R/6$ away from the hole, as it must.

§6.2

Motion of the centre of mass

Differentiate $M\vec{r}_{\text{cm}} = \sum m_i \vec{r}_i$ twice with respect to time:

$$M\vec{a}_{\text{cm}} = \sum m_i \vec{a}_i = \sum \vec{F}_i.$$

The force on each particle is partly external and partly internal, exerted by the other particles of the system. By Newton's third law, internal forces come in equal and opposite pairs, so they cancel in the sum. What remains is:

KEY IDEA

$$\sum \vec{F}_{\text{ext}} = M\vec{a}_{\text{cm}}.$$

The centre of mass of any system moves like a single particle of mass M acted on by the net **external** force. Internal forces, however large, cannot change the motion of the CM.

This is why we could treat cars, people and planets as particles in earlier chapters. Their CM really does obey $\vec{F} = m\vec{a}$, whatever their parts are doing.

WORKED EXAMPLE 6.4

An exploding shell

A shell is fired from level ground and would land 400 m away. At the top of its flight it explodes into two fragments of equal mass. One fragment comes momentarily to rest and falls straight down. Where does the other land? Ignore air resistance.

Solution

Key idea. The explosion's forces are internal, so the CM continues along the original parabola and lands 400 m away. (This holds as long as both fragments land at the same time, which they do here: both start the fall from the same height with zero vertical velocity.)

The top of the flight is at $x = 200$ m, so fragment 1 lands at $x_1 = 200$ m. For equal masses:

$$x_{\text{cm}} = \frac{x_1 + x_2}{2} = 400 \text{ m} \Rightarrow x_2 = 800 - 200 = 600 \text{ m}.$$

Check with momentum. Just before the explosion, the shell moves horizontally at v_x . Just after, fragment 1 has zero velocity, so fragment 2 must have $2v_x$ to conserve momentum:

$Mv_x = \frac{M}{2}(2v_x)$. Moving at twice the speed for the same falling time, it covers 400 m from the top instead of 200 m, landing at $200 + 400 = 600$ m. ✓

§6.3

Linear momentum and impulse

The **linear momentum** of a particle is $\vec{p} = m\vec{v}$, a vector measured in kg m/s. Newton's second law in its general form is

$$\vec{F}_{\text{net}} = \frac{d\vec{p}}{dt}.$$

Integrate this over the time a force acts. The **impulse** of the force is

$$\vec{J} = \int_{t_i}^{t_f} \vec{F} dt,$$

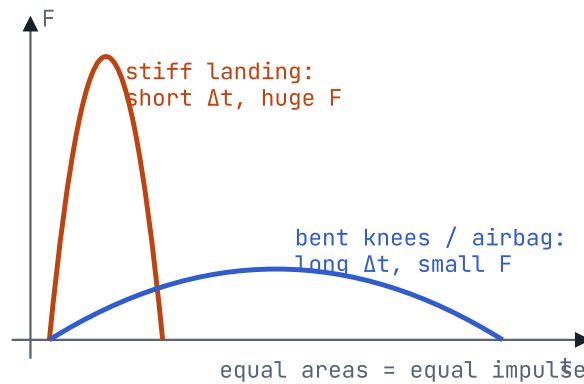
which is the area under the $\vec{F}$ - t graph.

KEY IDEA

Impulse–momentum theorem.

$$\vec{J}_{\text{net}} = \Delta\vec{p} = \vec{p}_f - \vec{p}_i.$$

For a collision lasting Δt , the average force is $\vec{F}_{\text{avg}} = \Delta\vec{p}/\Delta t$. For a given change in momentum, a longer stopping time means a smaller force.



The same impulse (the same change in momentum) delivered over different times. Cushioning spreads the impulse out and cuts the peak force.

WORKED EXAMPLE 6.5

Hitting a baseball

A 0.15 kg baseball arrives at 40 m/s and leaves the bat at 50 m/s in the opposite direction. The contact lasts 1.5 ms. Find the impulse and the average force on the ball.

Solution

Take the outgoing direction as positive, so $v_i = -40$ m/s and $v_f = +50$ m/s.

$$J = m(v_f - v_i) = 0.15(50 - (-40)) = 0.15(90) = 13.5 \text{ N}\cdot\text{s}.$$

$$F_{\text{avg}} = \frac{J}{\Delta t} = \frac{13.5}{1.5 \times 10^{-3}} = 9.0 \times 10^3 \text{ N}.$$

Evaluate. The average force is about 6000 times the ball's weight. The peak force is roughly twice the average. Reversing the direction makes $|\Delta v| = 90$ m/s, not 10 m/s: a very common sign error.

WORKED EXAMPLE 6.6

Landing from a jump

A 70 kg person jumps down from a wall 1.0 m high. Estimate the average force on their legs if they land (a) stiff-legged, stopping in about 1.0 cm, and (b) bending their knees, stopping in about 50 cm.

Solution

Impact speed: $v = \sqrt{2gh} = \sqrt{19.6} = 4.43$ m/s. The change in momentum is the same in both cases: $\Delta p = 70(4.43) = 310$ N s.

Assume a constant deceleration over the stopping distance d . Then $a = v^2/2d$, and the ground must supply $N = m(g + a)$.

(a) $d = 0.010$ m: $a = 19.6/0.020 = 980$ m/s² (100 g), so $N = 70(990) \approx 6.9 \times 10^4$ N, about 100 body weights. The stop takes $\Delta t = 2d/v = 4.5$ ms.

(b) $d = 0.50$ m: $a = 19.6$ m/s², so $N = 70(29.4) = 2.1 \times 10^3$ N, about 3 body weights. The stop takes 0.23 s.

Evaluate. The impulse is identical in both cases, but bending the knees stretches the stopping time about 50-fold and cuts the force by the same factor. Forces of tens of kilonewtons can fracture the tibia or calcaneus (heel bone), which is why falls from modest heights onto stiff legs cause serious injuries. The same physics explains crumple zones, airbags, helmets, running shoes, and gymnasts' landing mats.

§6.4

Conservation of linear momentum

For a system of particles, the total momentum is $\vec{P} = \sum \vec{p}_i = M\vec{v}_{\text{cm}}$, and

$$\frac{d\vec{P}}{dt} = \sum \vec{F}_{\text{ext}}.$$

KEY IDEA

Conservation of momentum. If the net external force on a system is zero, its total momentum is constant:

$$\sum \vec{p}_{\text{before}} = \sum \vec{p}_{\text{after}}.$$

Each component is conserved separately. If only the horizontal external force is zero, horizontal momentum is conserved even when vertical momentum is not.

In a short collision or explosion, external forces such as gravity and friction are usually tiny compared with the internal impact forces, and they act only for a very short time. Momentum is then *very nearly* conserved during the event, even if it is not over longer times.

WORKED EXAMPLE 6.7

Recoil, and walking on a boat

(a) A 4.0 kg rifle fires a 10 g bullet at 800 m/s. Find the rifle's recoil speed. (b) A 70 kg person stands at one end of a 140 kg boat, 4.0 m long, floating at rest in still water. The person walks to the other end. How far does the boat move? Ignore water resistance.

Solution

(a) The initial momentum is zero:

$$0 = m_b v_b + m_r v_r \Rightarrow v_r = -\frac{0.010(800)}{4.0} = -2.0 \text{ m/s}.$$

The rifle recoils at 2.0 m/s.

(b) No horizontal external force acts, and the system starts at rest, so the **CM cannot move**. Let the boat move a distance d backward. The person then moves $4.0 - d$ forward relative to the water. Keeping the CM fixed:

$$m_p(4.0 - d) = m_b d \Rightarrow d = \frac{m_p(4.0)}{m_p + m_b} = \frac{70(4.0)}{210} = 1.33 \text{ m}.$$

Evaluate. The person moves 2.67 m relative to the water and the boat 1.33 m the other way. The ratio of distances is the inverse ratio of the masses. Anyone stepping off a small boat onto a dock has felt this, as the boat slides away.

§6.5

Collisions

In every collision between isolated bodies, **momentum is conserved**. Kinetic energy may or may not be conserved:

TYPE	KINETIC ENERGY	EXAMPLE
Elastic	Conserved	Billiard balls (nearly), atoms and molecules, neutron scattering
Inelastic	Some lost, to heat, sound and deformation	Most real collisions
Perfectly inelastic	Maximum possible loss: the bodies stick together	Clay, car crashes where vehicles lock, ballistic pendulum

The **coefficient of restitution** measures how “bouncy” a head-on collision is:

$$e = \frac{\text{speed of separation}}{\text{speed of approach}} = \frac{v'_2 - v'_1}{v_1 - v_2}.$$

It ranges from $e = 1$ (elastic) to $e = 0$ (perfectly inelastic).

Elastic collisions in one dimension

Let m_1 moving at v_1 hit m_2 at rest, head-on and elastically. Two conservation laws apply:

$$m_1 v_1 = m_1 v'_1 + m_2 v'_2, \quad \frac{1}{2} m_1 v_1^2 = \frac{1}{2} m_1 v'^2_1 + \frac{1}{2} m_2 v'^2_2.$$

Rewrite them as $m_1(v_1 - v'_1) = m_2 v'_2$ and $m_1(v_1^2 - v'^2_1) = m_2 v'^2_2$. Divide the second by the first: $v_1 + v'_1 = v'_2$. This says the relative velocity simply reverses ($e = 1$). Solving:

$$v'_1 = \frac{m_1 - m_2}{m_1 + m_2} v_1, \quad v'_2 = \frac{2m_1}{m_1 + m_2} v_1.$$

Special cases:

- **Equal masses:** $v'_1 = 0$ and $v'_2 = v_1$. The bodies swap velocities (Newton’s cradle, a billiard ball’s “stop shot”).

- Light hits very heavy ($m_1 \ll m_2$): $v'_1 \approx -v_1$, and the light body bounces straight back. A ball bouncing off a wall is the familiar example.
- Heavy hits very light ($m_1 \gg m_2$): $v'_2 \approx 2v_1$. A golf club drives the ball off at nearly twice the club-head speed.

WORKED EXAMPLE 6.8

Slowing neutrons in a nuclear reactor

Fission produces fast neutrons, but they cause further fission efficiently only after being slowed down. This is done by elastic collisions with the nuclei of a *moderator*. What fraction of its kinetic energy does a neutron lose in a head-on elastic collision with (a) a carbon-12 nucleus (graphite) and (b) a hydrogen nucleus (a proton, in water)?

Solution

The neutron's kinetic energy after the collision is a fraction $(v'_1/v_1)^2$ of its initial value, so it keeps

$$\frac{K'_1}{K_1} = \left(\frac{m_1 - m_2}{m_1 + m_2} \right)^2.$$

(a) Carbon, with $m_2 = 12m_1$: $K'_1/K_1 = (11/13)^2 = 0.716$. The neutron loses 28% of its energy per head-on collision.

(b) Hydrogen, with $m_2 \approx m_1$: $K'_1/K_1 = 0$. The neutron can lose **all** its energy in a single head-on collision.

Evaluate. The best moderators are made of light nuclei. Water is an excellent moderator, but hydrogen also absorbs some neutrons. Heavy water and graphite absorb fewer, which is why they were used in early reactors. The same physics explains why hydrogen-rich materials (water, polyethylene, concrete) are used for neutron shielding around radiotherapy and research facilities.

Perfectly inelastic collisions

When the bodies stick together:

$$m_1v_1 + m_2v_2 = (m_1 + m_2)V.$$

The kinetic energy lost is as large as momentum conservation allows. All of the kinetic energy

measured in the centre-of-mass frame is lost. The kinetic energy of the CM motion, $\frac{1}{2}(m_1 + m_2)V^2$, must remain, because momentum must be conserved.

WORKED EXAMPLE 6.9

The ballistic pendulum

A 10 g bullet is fired into a 2.0 kg wooden block hanging from light cords. The block, with the bullet embedded, swings up until it is 10 cm higher than its starting position. Find the bullet's speed, and the fraction of its kinetic energy that was lost in the impact.

Solution

Two stages, two principles.

Stage 1, the collision (momentum is conserved, energy is not):

$$mv = (m + M)V.$$

Stage 2, the swing (mechanical energy is conserved, momentum is not, because the cords exert an external force):

$$\frac{1}{2}(m + M)V^2 = (m + M)gh \Rightarrow V = \sqrt{2gh} = \sqrt{2(9.8)(0.10)} = 1.40 \text{ m/s}.$$

Combining the two stages:

$$v = \frac{m + M}{m}V = \frac{2.01}{0.010}(1.40) = 281 \text{ m/s}.$$

Fraction of KE lost in the collision:

$$\frac{K_{\text{after}}}{K_{\text{before}}} = \frac{\frac{1}{2}(m + M)V^2}{\frac{1}{2}mv^2} = \frac{m}{m + M} = \frac{0.010}{2.01} = 0.005.$$

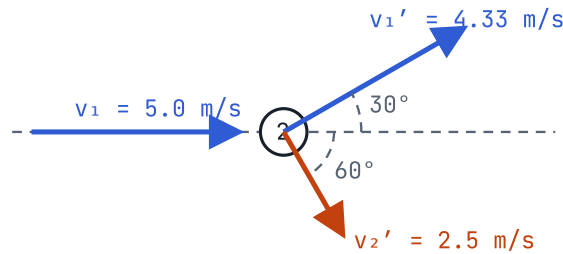
So 99.5% of the bullet's kinetic energy becomes heat and deformation.

Evaluate. It would be wrong to set $\frac{1}{2}mv^2 = (m + M)gh$. Energy is not conserved in the collision. Always identify which law applies to which stage.

Collisions in two dimensions

In two dimensions, conserve **each component** of momentum separately. For an elastic collision, also conserve kinetic energy. A useful result: when two **equal masses** collide elastically and one was initially at rest, they move off at **right angles** to each other (unless the collision is head-on). To

prove it, square the momentum equation $\vec{v}_1 = \vec{v}_1' + \vec{v}_2'$ and compare it with energy conservation, $v_1^2 = v_1'^2 + v_2'^2$. The cross term must vanish, so $\vec{v}_1' \cdot \vec{v}_2' = 0$.



An elastic glancing collision between equal masses. The two final velocities are perpendicular (Example 6.10).

WORKED EXAMPLE 6.10

A glancing collision of two pucks

A hockey puck moving at 5.0 m/s strikes an identical puck at rest. The collision is elastic, and the first puck is deflected 30° from its original direction. Find both final velocities.

Solution

Set up. Take x along the original motion. By the right-angle rule, puck 2 must move off at 60° on the other side of the x -axis.

Conserve momentum (the masses cancel):

$$x : 5.0 = v_1' \cos 30^\circ + v_2' \cos 60^\circ, \quad y : 0 = v_1' \sin 30^\circ - v_2' \sin 60^\circ.$$

From the y equation: $v_2' = v_1' \sin 30^\circ / \sin 60^\circ = 0.577 v_1'$. Substitute into the x equation:

$$5.0 = v_1'(0.866 + 0.577 \times 0.5) = 1.155 v_1' \Rightarrow v_1' = 4.33 \text{ m/s}, \quad v_2' = 2.50 \text{ m/s}.$$

Check kinetic energy (per unit mass, doubled):

$$4.33^2 + 2.50^2 = 18.75 + 6.25 = 25.0 = 5.0^2. \checkmark$$

These are just $v_1' = v_1 \cos 30^\circ$ and $v_2' = v_1 \sin 30^\circ$, as the right-angle geometry predicts.

§6.6

Systems of variable mass: rockets

A rocket gains speed by expelling exhaust backwards at speed u relative to itself. At time t , let the rocket have mass m and velocity v . In a short time dt it expels a mass dm_e of exhaust. The rocket's mass becomes $m - dm_e$ and its velocity $v + dv$, while the exhaust moves at $v - u$. With no external force, momentum is conserved:

$$mv = (m - dm_e)(v + dv) + dm_e(v - u).$$

Expanding and dropping the tiny product $dm_e dv$ gives $m dv = u dm_e$. Since the rocket's mass decreases, $dm = -dm_e$, so

$$m dv = -u dm.$$

Integrate from the initial mass m_0 (at speed v_0) to the final mass m_f :

$$\Delta v = v_f - v_0 = u \ln \frac{m_0}{m_f}.$$

KEY IDEA

The rocket equation (Tsiolkovsky). $\Delta v = u \ln(m_0/m_f)$. The thrust is $F = u |dm/dt|$. The final speed depends on the exhaust speed and the *mass ratio*, not on how quickly the fuel is burnt (apart from the time gravity acts during a vertical launch).

WORKED EXAMPLE 6.11

Why rockets have stages

A chemical rocket has an exhaust speed of 3.0 km/s. What mass ratio would a single-stage rocket need to reach 9.4 km/s, the Δv needed for low Earth orbit once gravity and drag losses are included?

Solution

$$\frac{m_0}{m_f} = e^{\Delta v/u} = e^{9.4/3.0} = e^{3.13} = 23.$$

The rocket would have to be about 96% fuel at launch, with only 4% left for the tanks, engines, structure *and* payload.

Evaluate. That is barely possible to build. **Staging** solves the problem: drop empty tanks and engines along the way, so that later stages do not waste fuel accelerating dead weight. A two-stage rocket in which each stage has a mass ratio of 5 gives $\Delta v = 2u \ln 5 = 9.7$ km/s. Because of the logarithm, raising the exhaust speed (better fuels, ion engines) is worth far more than adding fuel.

IN PRACTICE

Momentum in medicine and engineering.

- **Ballistocardiography.** Each heartbeat ejects about 70 g of blood towards the head. By momentum conservation, the body recoils very slightly the other way. Sensitive beds and wearable sensors detect this recoil to monitor cardiac output without electrodes (Problem P6.11).
- **Gait analysis.** Force plates in biomechanics labs record $F(t)$ under the foot. The impulse gives the change in the body's momentum, which is used to assess athletes, prosthetic limbs and patients recovering from injury.
- **Crash testing.** Engineers design structures to stretch out Δt and lower the peak force. They also design the momentum transfer between vehicles of different masses, which is why a small car fares worse in a collision with a heavy one.
- **Jet propulsion in nature.** Squid and jellyfish swim by expelling water, a living rocket.

CHAPTER SUMMARY

- Centre of mass: $\vec{r}_{\text{cm}} = \sum m_i \vec{r}_i / M$, or $\int \vec{r} dm / M$. Use symmetry, and treat holes as negative masses.
- $\sum \vec{F}_{\text{ext}} = M \vec{a}_{\text{cm}}$. Internal forces cannot move the CM.
- Momentum $\vec{p} = m \vec{v}$. Impulse $\vec{J} = \int \vec{F} dt = \Delta \vec{p}$. A longer contact time means a smaller force.
- If $\sum \vec{F}_{\text{ext}} = 0$, the total momentum is conserved, component by component.
- Elastic collisions conserve kinetic energy. Head-on with m_2 at rest: $v'_1 = \frac{m_1 - m_2}{m_1 + m_2} v_1$ and $v'_2 = \frac{2m_1}{m_1 + m_2} v_1$.
- Perfectly inelastic: the bodies stick, $V = \sum m_i v_i / \sum m_i$. Coefficient of restitution: $e = \text{separation speed} \div \text{approach speed}$.
- Rocket equation: $\Delta v = u \ln(m_0/m_f)$. Thrust = $u |dm/dt|$.

Practice problems

Full step-by-step solutions are in the separate solutions PDF.

LEVEL A – CONCEPT CHECK

P6.1

Can the centre of mass of a body lie outside the body? Give two examples. Explain how a high jumper using the “Fosbury flop” can clear the bar while their centre of mass passes *under* it.

P6.2

Explain, using the impulse–momentum theorem, why you bend your knees when landing from a jump, and why an egg dropped onto a pillow does not break while one dropped onto a floor does.

P6.3

In a perfectly inelastic collision, is all the kinetic energy lost? Explain, referring to the motion of the centre of mass.

P6.4

An astronaut drifts at rest 10 m from her spacecraft, her safety line broken. She is holding a heavy wrench. How can she get back? Can she get back by “swimming” motions alone?

LEVEL B – STANDARD PROBLEMS**P6.5**

A uniform thin plate is made of three identical squares of side 10 cm, arranged in an L shape. Taking the corner of the L as the origin, the squares occupy $0 \leq x, y \leq 10$, then $10 \leq x \leq 20$ with $0 \leq y \leq 10$, then $0 \leq x \leq 10$ with $10 \leq y \leq 20$ (all in cm). Find the centre of mass. Does it lie on the plate?

P6.6

The masses of the Earth and Moon are 5.97×10^{24} kg and 7.35×10^{22} kg, and their centres are 3.84×10^8 m apart. Locate the centre of mass of the Earth–Moon system. Is it inside the Earth (radius 6.37×10^6 m)? What does this mean for the Earth’s motion?

P6.7

A 0.40 kg ball strikes a wall perpendicularly at 12 m/s and rebounds at 10 m/s. Contact lasts 20 ms. Find the impulse on the ball and the average force the wall exerts. What fraction of the ball’s kinetic energy is lost?

P6.8

A 58 g tennis ball, initially at rest, is struck by a racket. The force rises linearly from zero to a peak of 600 N in 5.0 ms, then falls linearly back to zero in another 5.0 ms. Find the impulse and the ball’s speed as it leaves the racket.

P6.9

A 2.0 kg cart moving right at 3.0 m/s collides with a 1.0 kg cart moving left at 2.0 m/s, and they stick together. Find their common velocity and the kinetic energy lost.

P6.10

A 1.0 kg ball moving at 6.0 m/s collides head-on and elastically with a 2.0 kg ball at rest. Find both velocities after the collision, and verify that kinetic energy is conserved.

P6.11

Ballistocardiography. During ventricular ejection, about 70 g of blood is accelerated from rest to about 1.0 m/s, towards the head, in roughly 0.10 s. A 70 kg patient lies on a nearly frictionless air-bearing table. (a) Find the patient's recoil velocity and its direction. (b) Find the average recoil force on the body. (c) Why is a ballistocardiograph signal so small, and why must the bed be nearly frictionless to measure it?

P6.12

A 4.0 kg rifle fires a 10 g bullet at 800 m/s. (a) Find the recoil speed and the recoil kinetic energy, and compare it with the bullet's kinetic energy. (b) If the shooter's shoulder stops the rifle over 5.0 cm, find the average force on the shoulder. (c) Why should the rifle be held firmly against the shoulder rather than a few centimetres away from it?

LEVEL C – CHALLENGE PROBLEMS**P6.13**

Neutron moderation. A fission neutron starts with a kinetic energy of 2.0 MeV and must be slowed to a thermal energy of about 0.025 eV. (a) Assuming every collision with carbon-12 is head-on and elastic, how many collisions are needed? (b) In reality the collisions are at random angles, and on average a neutron keeps a fraction $e^{-\xi}$ of its energy per collision, where $\xi \approx 0.158$ for carbon and $\xi \approx 1.0$ for hydrogen. Estimate the number of collisions needed in graphite and in water.

P6.14

A rocket of initial mass 50 000 kg, including 40 000 kg of fuel, burns its fuel at a constant rate over 100 s. The exhaust speed is 2500 m/s. It is launched vertically from rest. Neglect air resistance and the variation of g with height. (a) Find the thrust, and the rocket's acceleration at lift-off. (b) Find the speed at burnout. Show that in a uniform gravitational field, $\Delta v = u \ln(m_0/m_f) - gt_b$. (c) Find the acceleration just before burnout.

P6.15

Sand falls vertically from a hopper onto a horizontal conveyor belt at a rate of 20 kg/s. The belt moves at a constant 2.0 m/s. (a) What horizontal force must the motor apply to keep the belt moving at constant speed? (b) What power does the motor deliver? (c) At what rate does the sand gain kinetic energy? Account for the difference.

P6.16

Two blocks of mass 1.0 kg and 3.0 kg sit on a frictionless surface with a light, compressed spring between them, held by a thread. The spring stores 24 J. The thread is burnt. Find the speed of each block after it leaves the spring. What fraction of the energy does each block receive?

P6.17

A ball is dropped from height h onto a hard floor. Its coefficient of restitution with the floor is e . (a) Show that after each bounce it rises to e^2 times its previous height. (b) Show that the total distance it travels before coming to rest is $h(1 + e^2)/(1 - e^2)$, and the total time is $\sqrt{2h/g}(1 + e)/(1 - e)$. (c) Evaluate both for $h = 10$ m and $e = 0.80$.

P6.18

A stationary bomb explodes into three fragments. A 1.0 kg fragment flies north at 30 m/s, and a 2.0 kg fragment flies east at 20 m/s. The third fragment has a mass of 3.0 kg. Find its velocity (magnitude and direction), and the total kinetic energy released by the explosion.

All worked examples and practice problems in these notes are original.

Textbooks for further reading and practice, with a chapter-by-chapter map, are listed in the References.

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